

hence the energy of the photon may not be the same.

- 7 a Newton's second law states that the rate of change of momentum of a body is directly proportional to the net force acting on it

B1

and (the momentum change) occurs in the same direction as the net force.

B1

b
$$\frac{dm}{dt} = \frac{F}{v} = \frac{34.7 \times 10^6}{2.6 \times 10^3}$$

M1

$$\frac{dm}{dt} = 1.33 \times 10^4 \text{ kg s}^{-1}$$

A1

ci
$$\frac{\Delta v_r}{V} = \ln\left(\frac{m_o}{m}\right)$$

$$-\left(\frac{\Delta v_r}{V}\right) = -\ln\left(\frac{m_o}{m}\right) \quad (\text{multiply both sides by } -1)$$

$$-\left(\frac{\Delta v_r}{V}\right) = \ln\left(\frac{m_o}{m}\right)^{-1}$$

$$-\left(\frac{\Delta v_r}{V}\right) = \ln\left(\frac{m}{m_o}\right)$$

M1

$$\left(\frac{m}{m_o}\right) = e^{-\left(\frac{\Delta v_r}{V}\right)}$$

A0

Working line shown and clear conversion of natural logs to exponentials.

cii $\frac{m}{m_o} = 0.88$

B1

$$\Delta v_r = 7740 \text{ m s}^{-1}$$

$\Rightarrow$ to list as 7.7 (on table as the header bears the power of tens extension)

B1

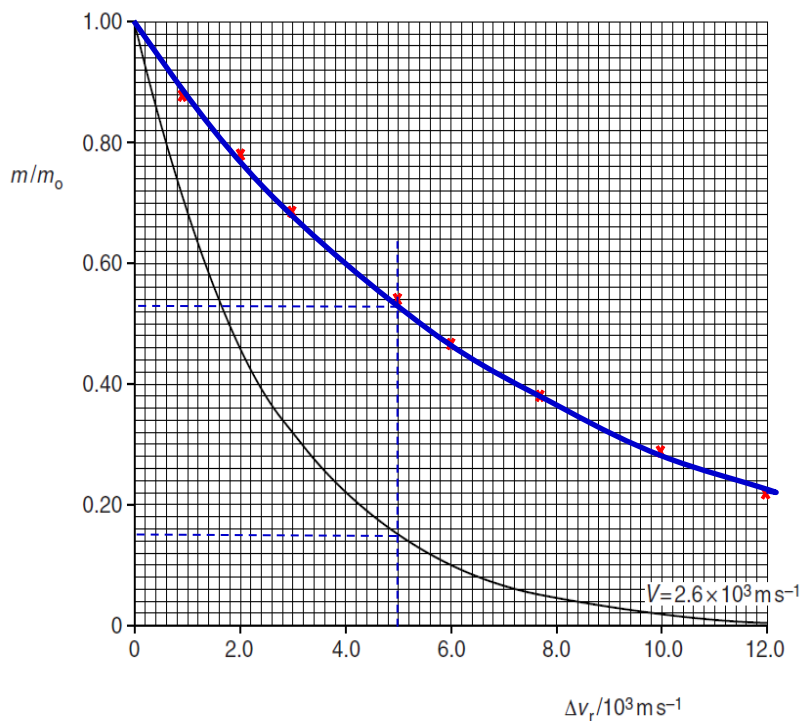
ciii 8 points correctly plotted (ECF the table values)

B2

One mark lost for each error, minimum of zero (precision of 0.5 small square)

Best fit smooth curve drawn

B1



civ same change of velocity, $\Delta v_r = 5.0 \times 10^3 \text{ m s}^{-1}$.

When $V = 2.6 \times 10^3$, $(m/m_o) = 0.15$

C1

$$m = (0.15)(2.04 \times 10^6) = 3.06 \times 10^5 \text{ kg}$$

When $V = 8.0 \times 10^3$, $(m/m_o) = 0.53$

C1

$$m = (0.53)(2.04 \times 10^6) = 1.08 \times 10^6 \text{ kg}$$

Difference in mass = 775 000 kg (allow ECF)

A1

Deduct 1 mark for each read off precision error of more than 0.5 small square.

di $E = -\frac{GM_E m_S}{(R+h)}$ **or** $-m_S v^2$ **B1**

No mark awarded if omit the negative sign

dii The gravitational potential energy is the amount of work done by an external agent **M1**
 in transferring the satellite from infinity to the point in question without a change in kinetic energy. **A1**

diii $KE = \frac{1}{2}mv_S^2 = \frac{1}{2}(152)(7.70 \times 10^3)^2 = 4.50 \times 10^9 \text{ J}$ **C1**

$GPE = \text{Total } E - KE = -4.5 \times 10^9 - 4.50 \times 10^9 = -9.0 \times 10^9 \text{ J}$ **C1**

$$E = -\frac{GM_E m_S}{(R+h)}$$

$$-9.0 \times 10^9 = -\frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})(152)}{(6.36 \times 10^6 + h)}$$
 M1

$h = 3.76 \times 10^5 \text{ m}$ **A1**

**Award marks for $GPE = 2 E_T$*

Alternative method

The gravitational force provides the centripetal force

$$\frac{GM_E m}{(R+h)^2} = \frac{mv^2}{R+h}$$